



REVA Academy for Corporate Excellence

Program: PGD/MTech/MS in Artificial Intelligence	Course: Computer Vision and Image Analytics
Batch: AI09	Exam: Module End Exam
Date: 27 th September 2025	Time: 2:00pm to 5:00pm
Duration: 3 Hrs.	Marks: 100

Instructions	
1.	This is an open-book exam. However, all the answers must be paraphrased and customized. Any evidence of copying verbatim from any third-party resources or plagiarized content will lead to suspension and other stringent actions. If any identical answers are found, both will be awarded zero.
2.	You may use any tool or software to answer the questions. All the outputs and codebase must be well commented.
3.	All the answers must be written in the answer script template provided by RACE. The template will be available in the LMS with the answer submission link. Along with the main answer script, all other artifacts like screenshots, codebase and any other materials must be submitted. All such artefacts must be named with the corresponding question number.
4.	All the artefacts must be zipped and submitted through the LMS Link. Zipped Folder must be named as 'SRN_Name of the Course'.

Instructions		Answer ONE FULL question from each unit.	Marks
UNIT - I			
Q1.	a.	Write Python code using OpenCV to load an image and convert it into Grayscale, HSV, and LAB color spaces . Attach the code, screenshots of outputs, and explain in your own words the visual differences.	5
	b.	A company wants to preprocess scanned invoices before applying OCR. Design a pipeline in Python to: • Resize input images to a fixed resolution • Denoise using Gaussian blur • Apply adaptive thresholding for better text extraction. Provide code, intermediate results, and justify why each step is necessary. You can download and use sample invoices from the web for this question.	10
	c.	Imagine you are building a photo editor tool for social media. Implement a script in OpenCV that supports crop, rotate, flip, and adding text/logo watermark. Submit the	10

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		code, example screenshots, and discuss how each operation could enhance real-world applications.		
OR				
Q2.	a.	Using a sample image of your choice (nature/portrait/product), generate and plot the histogram of pixel intensities in grayscale. Interpret what the histogram tells about brightness and contrast distribution.	5	
	b.	In manufacturing quality checks, filters are often used to detect defects. Write Python code to apply Gaussian blur, median filter, and sharpening filters on a product image (e.g., a circuit board). Compare the results and explain which filter best highlights fine defects and why.	10	
	c.	Develop a case study where edge detection (Sobel, Laplacian, Canny) is applied to an industry problem (e.g., detecting cracks in tiles or boundaries in satellite images). Implement code, visualize outputs, and critically evaluate which method is best suited.	10	
UNIT - II				
Q3.	a.	Use the Intel Image Classification dataset. (https://www.kaggle.com/datasets/puneet6060/intel-image-classification) (≈25k images, 6 categories like buildings, forest, street). Train a simple CNN classifier (3–4 convolution layers). Submit code, accuracy plots, and test results. In case the training is taking time, then take a small sample of images.	5	25
	b.	Compare your CNN with a pretrained ResNet18 (fine-tuned). Report accuracy, confusion matrix, and discuss why transfer learning works better on smaller datasets.	10	
	c.	Assume this model is deployed in a smart city monitoring system (classifying scenes from CCTV feeds). What risks exist if the model misclassifies (e.g., forest → building)? Suggest mitigation strategies.	10	
OR				

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Q4.	a.	Use the Sign Language MNIST dataset. (https://www.kaggle.com/datasets/datamunge/sign-language-mnist) (28x28 grayscale hand signs, 24 classes). Perform image clustering using: • Feature-based clustering (SIFT/ORB) OR • Deep embeddings from a pretrained CNN. • Visualize at least 2 clusters and explain results.		
	b.	Download any scanned text image from the web (e.g., an invoice, receipt, newspaper clipping, or old document). Apply Tesseract OCR to extract the text. • Provide the Python code used for OCR. • Show the original image and the extracted text output. • Highlight common errors (e.g., missing words, wrong characters, line breaks). • Analyze the likely causes of errors (blur, skew, handwriting, unusual fonts, poor image quality).	10	
	c.	Design a lightweight OCR + clustering pipeline for a school administration system (e.g., grouping student forms, reading names/IDs). Draw a workflow diagram, explain design choices, and evaluate industry benefits. You can take example forms from web.	10	
UNIT - III				
Q5.	a.	Download a few sample pedestrian images from the Penn-Fudan Pedestrian dataset (https://www.kaggle.com/datasets/psvishnu/pennfudan-database-for-pedestrian-detection-zip). Apply traditional object detection methods (HOG + SVM or Haar Cascade). Show bounding boxes on at least 5 images.	5	
	b.	Train a YOLOv8 small model (pretrained) on a subset of Penn-Fudan or COCO128 dataset. Run inference on test images, visualize bounding boxes, and report detection accuracy.	10	25
	c.	Compare results from traditional detection (HOG/Haar) vs YOLOv8. Discuss strengths, weaknesses, and real-world deployment considerations (e.g., speed vs accuracy).	10	

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	OR			
Q6.	a.	Using the Oxford-IIIT Pet Dataset (https://www.kaggle.com/datasets/tanlikesmath/the-oxfordiiit-pet-dataset), apply semantic segmentation with a pretrained DeepLabV3 model from TorchVision. Visualize segmentation masks for at least 5 images.	5	25
	b.	Perform instance segmentation using a pretrained Mask R-CNN model on the same images. Compare semantic vs instance masks (e.g., when multiple cats/dogs appear).	10	
	c.	Imagine deploying this system in a veterinary clinic to assist in analyzing pet scans/photos. Propose a workflow diagram, explain practical benefits, and highlight limitations of segmentation models in medical-like settings.	10	
	UNIT - IV			
Q7.	a.	Download any image from the web (e.g., a natural scene, an object). Use a pretrained Vision Transformer (ViT) from Hugging Face to classify the image. Show top 5 predicted classes with probabilities.	5	25
	b.	Using the same image, apply CLIP (Contrastive Language–Image Pretraining) with two different text prompts (e.g., “a photo of a dog” vs. “a photo of a cat”). Report similarity scores and analyze how CLIP bridges vision + language.	10	
	c.	Imagine building a visual search engine for an e-commerce site (customers search with text, system retrieves matching product images). Propose a pipeline using ViT + CLIP. Draw workflow diagram and critically evaluate potential challenges (e.g., bias, misclassification).	10	
	OR			
Q8.	a.	Use Stable Diffusion to generate an image from a creative text prompt (e.g., “ <i>A futuristic city skyline at sunset in watercolor style</i> ”). Save and display the output.	5	25
	b.	Modify the generated image using image-to-image generation (e.g., start with a real photo of a street and stylize it into cyberpunk theme). Provide code, inputs, and outputs.	10	

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	c.	Discuss a real-world application of diffusion models in industry (e.g., product design, marketing, healthcare imaging). Critically evaluate opportunities and ethical risks (e.g., misuse in deepfakes, copyright).	10	
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